National University of Computer and Emerging Sciences

Phase 2 Report: Data Acquisition & Setup

Attention-Guided Cross-Sensor Land Cover Classification

Group F26-06 | Track A (Research & Development)

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Repository: https://github.com/mahar3b/satelite-imagery-system-analysis

1. Dataset Source & Description

The project studies cross-sensor generalization: a model trained on Sentinel-2 imagery is tested on Landsat imagery of the same 10 land-cover classes. Two public datasets provide the source and target domains.

Property	EuroSAT (source domain)	EuroSAT-L (target domain)
Sensor	Sentinel-2, multispectral	Landsat (OLI + thermal)
Bands	13	8
Patch size	64x64 px at 10 m	22x22 px at 30 m
Ground footprint	~640 m	~660 m
Total images	27,000	27,000 (splits 16,200 / 5,400 / 5,400)
Where from	Hugging Face blanchon/EuroSAT_MSI (original: Zenodo 7711810; Helber et al., 2019)	Hugging Face isaaccorley/eurosat-l (Landsat-Bench, arXiv:2506.08780)
Format	Multispectral arrays, original value range	GeoTIFF, uint16, in a zip with split-list text files

Classes (shared): AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, SeaLake.

2. Data Collection Method

Both datasets come from public repositories, so no scraping or API work was needed (Track A, repository route).

- **Sentinel-2:** streamed with the Hugging Face datasets library. Only a stratified subset was kept (700 train and 200 test images per class) and only four bands (B02, B03, B04, B08), stored as compressed uint16 .npz files on Google Drive. This is about 230 MB instead of the full ~2 GB archive.
- **Landsat:** downloaded eurosat-l.zip and eurosat-test.txt directly from the Hugging Face repo. The 5,400-image test list defines the cross-sensor evaluation set.
- **Caching:** everything is saved to Drive, and the notebook skips downloads that already exist. Work runs on free Colab (T4 GPU) since no local GPU is available.
- **Issue met:** the landsatbench package's datamodule failed with an AttributeError, and the HF auto-converted version of EuroSAT-L only exposes filenames, not pixels. We therefore read the raw GeoTIFFs directly.

3. Preprocessing Steps

1. **Labels:** class is parsed from the filename (Forest_1399.jpg gives Forest). Split lists name .jpg files but the files on disk are .tif.
2. **Subsampling:** stratified, 700 train and 200 test images per Sentinel-2 class, to fit free-tier compute.
3. **Band selection:** Sentinel-2 keeps Blue, Green, Red, NIR. Landsat uses OLI bands 4, 3, 2 as RGB (array indices 3, 2, 1) and NIR at index 4; the thermal band (index 7) is ignored.

4. **Landsat contrast stretch:** raw reflectance is very dark (about 0.04 to 0.26), so each image is stretched between its 2nd and 98th percentile. A first attempt without this gave near-chance accuracy (12 to 16%), so the step is essential.
5. **Spectral indices:** NDVI = (NIR - Red) / (NIR + Red) and NDWI = (Green - NIR) / (Green + NIR), computed from Sentinel-2 bands.
6. **No-data handling:** some pixels have Red = 0, which pushes NDVI to exactly 1.0. These are masked when building NDVI/NDWI targets.
7. **Model input:** resize to 224x224 and ImageNet mean/std normalization.
8. **Integrity check:** 0 NaNs in the cached Sentinel-2 arrays; Landsat file counts per class sum to the 5,400-image split list.

4. Brief EDA (Key Insights)

4.1 Class balance and resolution

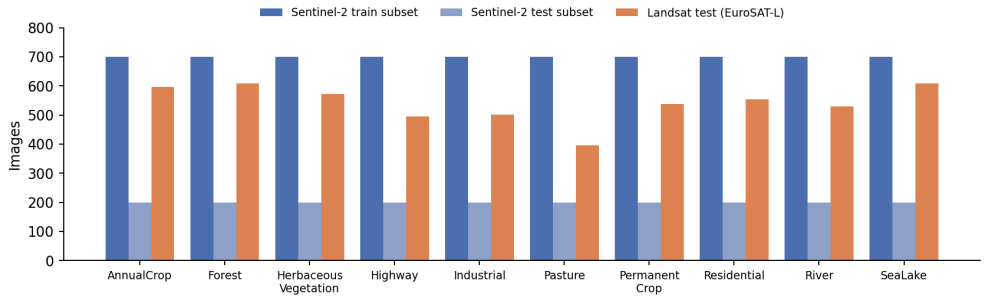


Figure 1. Images per class in each split.

- Sentinel-2 subsets are perfectly balanced. The Landsat test set is not: from 396 images (Pasture) to 609 (SeaLake). Per-class or macro metrics will be fairer than overall accuracy.
- For the same ~640 m of ground, Sentinel-2 gives 64x64 pixels and Landsat only 22x22. Small objects such as roads and rivers are much coarser in Landsat.

4.2 Does NDVI / NDWI separate the classes?

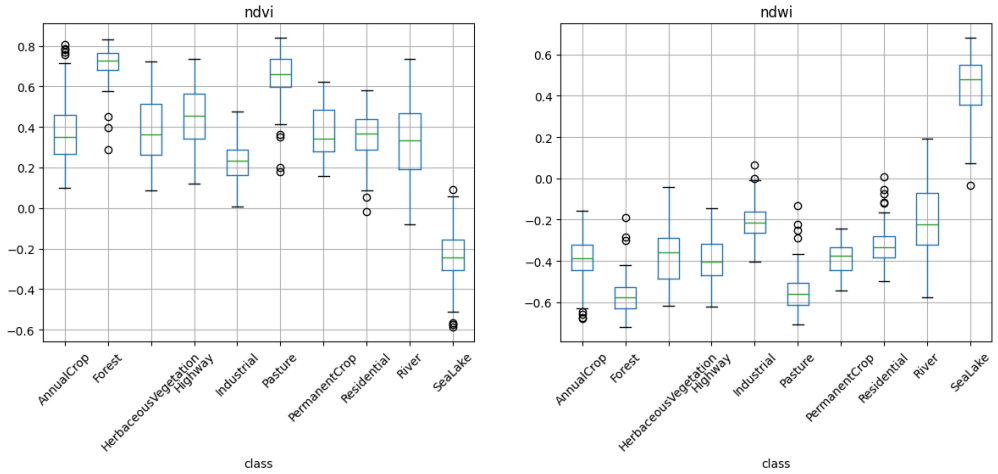


Figure 2. Per-image mean NDVI and NDWI by class (150 images per class).

- Forest (NDVI 0.717) and Pasture (0.654) stand out as high-vegetation classes. SeaLake is the only class with positive NDWI (0.451) and negative NDVI (-0.237).
- The other seven classes overlap heavily, with mean NDVI between 0.23 and 0.45. River (NDVI 0.331, NDWI -0.206) looks like a land class, because a 640 m patch is mostly surrounding vegetation.

4.3 Pixel-level structure

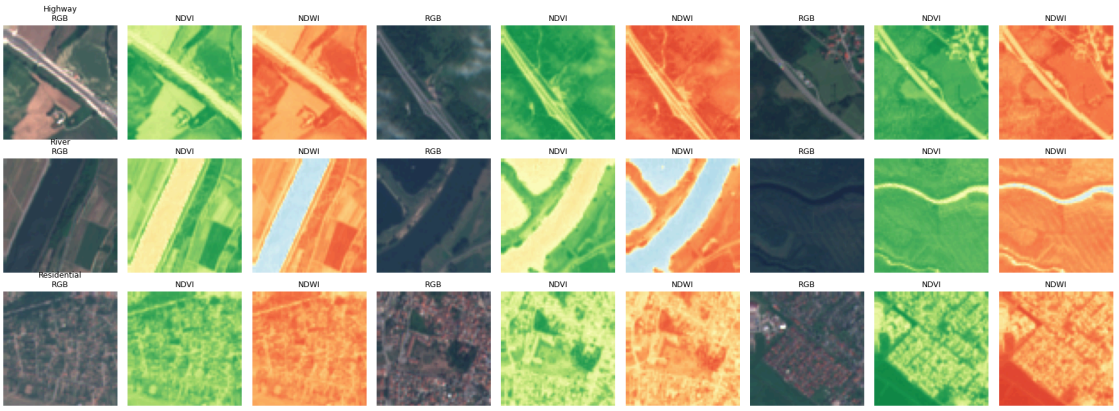


Figure 3. RGB, NDVI and NDWI for Highway, River and Residential patches (Sentinel-2).

- The whole-image averages hide real structure. In River patches, NDWI traces the channel sharply. In Highway patches, the road appears as a clear low-NDVI band.
- Residential patches are flat and textured with no clean structure, so dense built-up areas are the weak case for NDVI/NDWI supervision.

4.4 Landsat appearance

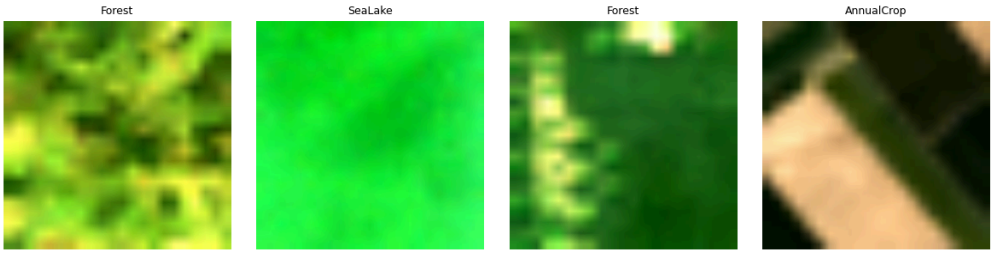


Figure 4. Landsat test samples after contrast stretching (Forest, SeaLake, Forest, AnnualCrop).

- Landsat patches show real field and tree-line structure but are blurry and have a strong green tint. The SeaLake sample shows no visible water, consistent with coarse resolution mixing shoreline vegetation into the patch.

5. Final Dataset Summary

Class	Sentinel-2 train	Sentinel-2 test	Landsat test
AnnualCrop	700	200	596
Forest	700	200	608
HerbaceousVegetation	700	200	573
Highway	700	200	496
Industrial	700	200	501
Pasture	700	200	396
PermanentCrop	700	200	538
Residential	700	200	554
River	700	200	529
SeaLake	700	200	609
Total	**7,000**	**2,000**	**5,400**

	Source (Sentinel-2)	Target (Landsat)
Images used	7,000 train + 2,000 test	5,400 test (evaluation only)
Channels kept	Blue, Green, Red, NIR	RGB (stretched); NIR available
Input to models	224x224, ImageNet normalized	224x224, ImageNet normalized
Stored in	Drive: s2_train.npz, s2_test.npz	Drive: eurosat-l.zip, eurosat-test.txt

The data is clean, labeled, cached and ready for model implementation. Known limitations: training uses a 7,000-image subset for compute reasons, the Landsat test set is mildly imbalanced, and NDVI/NDWI are weak supervision signals for dense built-up classes.